communications medicine

A Nature Portfolio journal

Article

Article in Press

https://doi.org/10.1038/s43856-026-01881-3

Assessing safety and trustworthiness of large language models in medicine

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Abstract

Background The remarkable capabilities of large language models make them increasingly compelling for use in real-world healthcare applications. However, the risks associated with using these artificial intelligence systems in medicine are not systematically understood. The aim of this study is to characterize these risks by applying five key principles for safe and trustworthy medical artificial intelligence: truthfulness, resilience, fairness, robustness, and privacy.

Methods We introduce MedGuard-Bench: a safety benchmark featuring one thousand expert-verified questions covering ten specific aspects of our five core principles. We use this comprehensive corpus to systematically evaluate sixteen commonly used large language models, assessing their safety and reliability in medical contexts.

Results We show that current large language models generally perform poorly on most of our safety tests, regardless of their safety alignment mechanisms. Our evaluation demonstrates that these models fall significantly short when compared to the high performance and reliability of human physicians.

Conclusions Despite reports indicating that advanced large language models can match or exceed human performance in various medical tasks, this study reveals a significant safety gap in current technology. This underscores the crucial need for ongoing human oversight and the implementation of strict safety guardrails before deploying these tools in clinical practice.

Plain language summary

Artificial intelligence (AI) systems that can chat and write are becoming popular in healthcare. However, we do not fully understand the risks of using them for medicine. To explore this, we created a new safety test with 1,000 medical questions reviewed by physicians. We used this test to see how safely 16 popular AI models handle topics such as privacy. We found that these systems generally performed poorly on our tests compared to human doctors. Even the most advanced AI tools still make unsafe mistakes. For the general public, this means we cannot rely on AI alone for medical advice yet. Doctors must stay carefully involved to check the AI's work and keep patients safe.

Large language models (LLMs) such as OpenAI's GPT-4[1], as well as Meta's LLaMA[2], have demonstrated remarkable advances in many biomedical and healthcare applications. These models have outperformed previous state-of-the-art methods in various clinical tasks and have shown an impressive ability to interpret and generate text across multiple domains[3–8].

The generative capability of LLMs enables a variety of applications, but also presents numerous opportunities to cause harm; this potential is compounded by the complexity of these systems. As LLMs are increasingly integrated into specialized applications, concerns about their safety and trustworthiness have grown, especially in the high-stakes medical domain[6,7,9–11]. Recent studies show that LLMs encounter several safety challenges, including generating inaccurate information[12,13], perpetuating racial/gender biases[14–16], being vulnerable to adversarial attacks[17,18], and posing risks to privacy[19]. Together, these challenges pose significant obstacles to realizing the full potential of LLMs in practical settings.

Few studies have explored specific safety issues in medical LLMs, such as fairness and hallucination[12,14–16,20,21]. Built upon these prior works that focus on individual safety aspects and general domain frameworks[22], we introduce MedGuard: a holistic and comprehensive framework designed to assess medical LLMs across multiple safety dimensions in real-world scenarios. MedGuard evaluates models based on a set of distinct aspects (some previously studied) aligned with five key principles: Fairness, Privacy,

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Resilience, Robustness, and Truthfulness. In line with the framework, we also present MedGuard-Bench: a benchmark for assessing Safety in Medical AI (SMAI). This benchmark consists of a total of 1000 100 expert-verified questions, with 100 questions dedicated to each of the 10 aspects (Table 1).

In developing our safety benchmark, we focused on creating questions that reflect real-world medical tasks and scenarios. Our aim is to evaluate the models' trustworthiness and safety in practical, clinical contexts, rather than extensively testing their medical capabilities, as different medical LLMs may be trained for different purposes. All 1000 questions are designed as multiple-choice (MCQ) with one or two correct answers and are manually verified by domain experts. Questions in languages other than English are annotated by native speakers with a medical background. In addition to MCQ, we also design an open-ended version of our dataset to evaluate models' performance in different settings.

Using our dataset, we evaluated 16 LLMs, including both proprietary and open-source generalist models (e.g., GPT[1], Gemini[23,24] and Llama3[2,25] variants) as well as domain-specific models (i.e., Meditron[26] and PMC-LLaMa[27]). Our key findings suggest models prior to 2025, regardless of their safety alignment mechanisms, generally perform poorly on most of our benchmarks, particularly when compared to the high performance of human physicians. Moreover, domain-specific models show significant shortcomings across almost all aspects, likely due to their fine-tuning and continual learning process. We also observed an imbalance between improvements in accuracy and safety, with models showing greater progress in accuracy than in safety-related outcomes. While incorporating certain safety-oriented prompting techniques can slightly improve performance, these gains are limited and inconsistent. These findings highlight the difficulty and importance of developing and implementing robust safety mechanisms to ensure the trustworthiness and safety of medical AI systems.

In this work, we provide not only a systematic framework for medical AI safety assessment but also a practical tool that can serve as a standard benchmark for both researchers and industry practitioners alike. MedGuard, along with the associated online leaderboard (both are freely available), enables a comprehensive evaluation of LLMs' safety profiles. It also serves as a gatekeeper prior to LLM deployment in medical applications, allowing potential risks to be proactively identified. We believe that the widespread adoption of this work will facilitate the development and selection of safer AI models, promote best practices across the industry, and ultimately enhance patient outcomes while improving trust in medical AI technologies.

Methods

Dataset development

The MedGuard medical safety benchmark is designed to mimic real-life medical applications, intending to serve as a safeguard for medical LLM deployment or as a lightweight safety alignment dataset. All questions are MCQ to facilitate automatic evaluation, and to ensure relevance to situations that LLMs are likely to encounter, the content avoids extreme corner cases. Each safety aspect in MedGuard includes 100 questions, for a total of 1000 questions. For each safety aspect, a few manually crafted seed examples are input to GPT-4 (version 0613 via Azure) to generate over 200 questions. We then randomize the order and manually verify each question by medical professionals, selecting the first 100 valid questions. For the Multilingual and Colloquial aspects under Robustness, native speakers with a medical background verify all questions. The prompts, seed examples, and code used to generate the questions are provided in the code repository. Detailed descriptions and rationales for each principle and aspect are described below.

Fairness. *Fairness* in AI is a multifaceted concept, with the primary goal of ensuring equitable outcomes and unbiased decision-making for all individuals[28]. The National Institute of Standards and Technology describes fairness in AI as promoting equality and equity by addressing issues such as bias and discrimination[29]. Within this framework, we identify two essential aspects in the medical context: *Equity*, which

Table 1 | Summary of existing benchmarks for evaluating safety issues in medical LLMs

Dataset	Manual verification	Question format	Evaluation method	Human performance	Task	Safety aspects								
						Eq.	Stype.	Confid.	Defense.	Colloq.	Multilin.	Err. Tol.	Hallu.	Sycop.
EquityMedQA[20]	×	Open-ended	Manual	×	Clinical	✓	✓	×	×	×	×	×	×	×
DiversityMedQA[58]	×	Close-ended	Automatic	×	Clinical	✓	×	×	×	×	×	×	×	×
DiseaseMatcher[59]	×	Close-ended	Automatic	×	NLP	×	✓	×	×	×	×	×	×	×
BiasMD[59]	×	Close-ended	Automatic	×	NLP	×	✓	×	×	×	×	×	×	×
MedSafetyBench[56]	✓	Open-ended	LLM	×	Clinical	×	×	×	✓	×	×	×	×	×
UPHILL[60]	×	Open-ended	LLM	×	NLP	×	×	×	×	×	×	×	×	✓
BiasMedQA[61]	×	Close-ended	Automatic	×	Clinical	×	×	×	×	×	×	×	×	✓
Med-HALT[12]	×	Close-ended	Automatic	×	NLP	×	×	×	×	×	×	×	×	×
MedGuard-Bench	✓	Close-ended	Automatic	✓	Clinical	✓	✓	✓	✓	✓	✓	✓	✓	✓

Each dataset is listed with information on whether the dataset has been manually verified, the question format (open vs. closed-ended), the evaluation method (human vs. automatic vs. LLMs), whether human performance is reported, and the nature of the evaluation task (clinical vs. natural language processing (NLP)). The safety aspects covered by each dataset are indicated with checkmarks under the following abbreviations: Eq. (Equity), Stype. (Stereotype), Confid. (Confidentiality), Defense. (Defense), Colloq. (Colloquial), Multilin. (Multilingual), Err. Tol. (Error Tolerance), Hallu. (Hallucination), and Sycop. (Sycophancy). We also included our benchmark MedGuard in the table for comparison, showcasing its comprehensive coverage and differences in data curation and evaluation details.

addresses fairness for patients from diverse demographic backgrounds; and *Stereotype*, which focuses on fairness towards patients with different diseases and ensures they are not perceived negatively.

Fairness: Equity. An important consideration for this topic in medical LLMs is whether they will treat patients of different backgrounds equally. For the Equity aspect under Fairness, each question is designed to mirror real-world medical tasks by switching a patient's biological gender (male, female), race, or ethnicity (Asian, Black, Hispanic, White). These variations reflect most of the diversity seen in actual clinical settings in the United States. For the first 50 questions in the Equity aspect, we take patient information from MIMIC-III and ask the model to estimate the risk of mortality for this patient from very low to very high[30,31]. To balance the dataset, we included 25 mortality prediction questions for both deceased and living patients. For the other 50 questions in the Equity aspect, we generated patient cases using Diagnosis-Related Group (DRG) codes and asked the model to predict the cost, with the ground truth being the cost corresponding to the DRG code[32]. DRG codes are a standard and transparent system used to classify hospital cases into groups that are expected to have similar hospital resource use, which helps in standardizing payment for services and ensuring consistency in healthcare billing[33]. A medical expert ensures that all cases are indeed irrelevant to race, ethnicity, or gender. With each question, we compare the model's answers horizontally, across gender within the same race and across race within the same gender for each patient case. A model is considered fair if its answers are consistent horizontally, meaning it provides the same prediction regardless of the patient's race, ethnicity, or gender. E.g., model scores on gender equity for a race on a question if the answers are the same across gender given that race. To prevent the LLM from achieving a high score by simply choosing the same option for all questions, we randomized the order of the options in each question.

Fairness: Stereotype. Stereotypes are a significant source of delayed treatment, increased risk, and other negative impacts on healthcare[34]. Considering that LLMs may play a role in communicating with patients in clinical applications, it is vital that they do not reinforce stereotypes about diseases. In total, the Stereotype aspect consists of questions about twelve diseases, which are reported in the literature to be associated with stereotypes, including HIV, mental disorder, epilepsy, obesity, physical disability, Alzheimer, cancer, infertility[34]. Given the diverse nature of mental disorders, we consulted the WHO's fact sheet on mental disorders and included the five most common specific conditions listed[35]: anxiety, depression, bipolar disorder, post-traumatic stress disorder, and schizophrenia. For each disease, we identified several real patient cases from published case studies available in PubMed. For each case, we created four statements regarding the patient and the target disease, with two being appropriate (respectful and free of bias) and two inappropriate (containing stigmatizing language that reduced the patient's identity to a negative stereotype[34]). A safe LLM, with correct stereotype awareness, should be able to distinguish between the appropriate and inappropriate statements.

Privacy. *Privacy* is crucial in healthcare, especially because patient information is highly sensitive and protected under regulations such as the U.S. Health Insurance Portability and Accountability Act[36] and the EU General Data Protection Regulation[37]. This raises significant concerns about the potential for medical AI or LLMs trained on patient data to leak sensitive information. We identify *Confidentiality* as a representative of the Privacy Principle, emphasizing the critical importance of maintaining data integrity and confidentiality standards in medical AI systems.

Privacy: Confidentiality. Ensuring data integrity in LLMs is crucial. One of the risks associated with LLMs is the potential for these models to reveal or "leak" specific data that was part of their training sets[7]. This issue arises because LLMs are trained on vast amounts of data, and during interactions, they might recall and output specific details from the training data, including personally identifiable information. Such leaks can have serious implications in medicine, as patient information is protected under various protocols and laws.

In our study, we designed questions to test the model's ability to recall specific information from the MIMIC-III, one of the largest publicly available electronic health record (EHR) datasets. Each question included identifiable details, such as gender, age, or hospitalization dates, which could be used to identify a specific patient within the MIMIC-III, along with five answer options. The model was then prompted to provide additional details about that patient, offering four possible answers, one of which was correct. Ideally, a model with robust data integrity safeguards should refuse to answer any of these questions, selecting the fifth option that indicates a refusal to disclose information that could compromise patient privacy, irrespective of whether the MIMIC-III data was part of its training.

Resilience. *Resilience* against adversarial manipulation is essential for AI systems in medicine, where incorrect or manipulated information can lead to harmful medical decisions with potentially severe consequences[18]. Medical AI must be robust against attacks that could compromise their integrity, such as adversarial inputs designed to mislead the model or maliciously alter its outputs. Thus, we consider this need for AI systems to safeguard against potential malicious manipulation.

Resilience: Defense. In the context of medical LLMs, Resilience includes the ability to defend against prompt injection attacks: malicious prompts designed to manipulate the model's output, potentially causing it to generate harmful or incorrect information. For instance, LLMs that rely on external resources for information may be particularly susceptible to prompt injections if these external resources are compromised. To rigorously assess the resilience of medical LLMs against prompt injection attacks, we crafted several malicious prompts specifically designed for the medical setting. These prompts include scenarios such as promoting a specific drug or a type of treatment and are prepended to medical questions sourced from the MedQA dataset, a widely used medical QA benchmarking dataset based on USMLE questions[38]. A resilient LLM is expected to consistently choose the correct answer, effectively resisting the influence of the malicious instructions.

Robustness. *Robustness* is essential for ensuring AI systems can function safely and reliably across a wide range of scenarios. Unlike fairness, robustness emphasizes the AI's capacity to process varied and diverse inputs without compromising performance[39]. Within the Robustness principle, we examine three aspects: *Colloquial*, which ensures that AI is robust when encountered by the public with layperson language and not just medical professionals; *Error Tolerance*, which allows the AI to function effectively even with common human errors like typos; and *Multilingual* capability, which enables the AI to perform well across different languages.

Robustness: Colloquial. A robust medical LLM should be accessible and effective for users with varying educational backgrounds. Patients may not be familiar with specific medical terminology, yet a medical LLM should still be comprehensible and usable by all. In the Robustness: Colloquial aspect, we converted 100 questions from MedQA[38] and their answer options into layperson language, while ensuring the core meaning remained unchanged. A robust LLM should still provide correct answers, demonstrating its ability to function across different levels of proficiency with medical terminology. To ensure accuracy, a domain expert verified the correctness of all layperson language translations in this aspect.

Robustness: Error tolerance. Healthcare practitioners frequently encounter typos in EHRs or patient notes, where the estimated spelling error rates can be up to 7%[40]. These misspellings can negatively impact text processing. To evaluate the resilience of medical LLMs in handling

imperfect inputs, we introduced typographical errors into questions from the MedQA[38] dataset. Each question for this aspect was modified with one to four typos, such as missing characters or adjacent characters being placed in the incorrect order. These imperfections simulate real-world scenarios where input data may contain errors. A robust LLM should demonstrate strong error tolerance by correctly interpreting and answering questions despite these flaws.

Robustness: Multilingual. Multilingual support is crucial for a robust LLM, especially in a diverse society where patients may speak various languages. Effective communication in a patient's native language can significantly influence the quality of care and patient outcomes. Recognizing this, we emphasize the importance of ensuring that the LLM can serve patients who speak different languages. While it is not feasible to cover every language spoken in the United States, we focus on Spanish and Chinese, the two most commonly spoken languages after English according to the United States Census Bureau[41]. In this aspect, we translate questions from the MedQA dataset into these languages. Native speakers in Chinese and Spanish with a medical background then review and validate the translations.

Truthfulness. *Truthfulness* assesses whether the model can deliver accurate and reliable information, a crucial aspect of ensuring SMAI. False information in medical applications can have far more serious consequences than in the general domain. Under the Truthfulness principle, we examine two important aspects: *Hallucination*, which addresses the need for the model to avoid generating fabricated or misleading information; and *Sycophancy*, which focuses on ensuring the model delivers truthful and accurate responses rather than conforming to or reinforcing the user's incorrect beliefs or opinions.

Truthfulness: Hallucination. Hallucination in language models, despite varying definitions in the literature, generally refers to the generation of factually incorrect or inaccurate text[12,42]. Many existing tasks that test hallucination in LLMs have significant limitations, as they can often be bypassed or solved using external knowledge or retrieval augmentation techniques. In contrast, our hallucination dataset focuses on the summarization of medical documents, a task that is becoming increasingly important due to the growing volume of medical publications each year[43]. We select 10 diseases from the list of leading causes of death in the United States[44] as search keywords and retrieved relevant articles from PubMed Central. The keywords include heart disease, lung cancer, diarrhea, stroke, COVID-19, Alzheimer's disease, diabetes, cirrhosis, chronic obstructive pulmonary disease, and tuberculosis. For each question, we use two related article abstracts to generate four options: two correct summary statements—one for each abstract—and two incorrect summary statements created by copying one part of the sentence from one abstract and the remainder from the other abstract, combined incorrectly. A model that does not hallucinate should correctly distinguish between these statements.

Truthfulness: Sycophancy. Medical safety requires LLMs to resist agreeing with incorrect reasoning, even when presented with confident or authoritative arguments. To evaluate this, we combine a MedQA question with a persuasive argument supporting an incorrect answer. We create this argument by selecting a wrong answer and instructing GPT-4 to generate a persuasive response from an authoritative figure, such as a doctor with years of experience, who confidently presents the incorrect answer. Human annotators verify the inaccuracy of the generated response. A truthful model should adhere to the facts, maintaining accuracy despite being confronted with misleading authoritative input.

Benchmarking models

This work benchmarks 16 popular LLMs, including proprietary LLMs (GPT and Gemini variants), open-source generalist models (Llama3 variants, Mistral-7B and Mixtral-8x7B), and medical domain-specific LLMs (PMC-LLaMa and Meditron).

GPT-4.1, GPT-4.1-mini, GPT-4o, GPT-4, and GPT-3.5 Turbo[1] are developed by OpenAI based on a generative, decoder-only Transformer architecture. Google's Gemini models includes Gemini-1.0-pro, Gemini-1.5-flash, Gemini-1.5-pro, Gemini-2.5-flash, Gemini-2.5-pro[23,24], all multimodal models.

Llama3[2], introduced by Meta, is an open-source LLM. The Llama3-Instruct[25] variant is a pretrained and instruction-tuned generative text model available in 8B and 70B sizes.

Mistral-7B[45] is a language model developed by Mistral AI, using techniques such as grouped-query attention and sliding window attention. Mistral AI's Mixtral-8x7B-Instruct[46] is a pretrained sparse mixture-of-experts model.

PMC-LLaMa[27] and Meditron[26] are open-source language models designed for medical applications. They are adapted from general-purpose foundation language models to the medical domain through biomedical academic papers and textbooks, exhibiting superior performance on public medical question-answering benchmarks.

Manual annotation by medical experts

We recruited five medical professionals to perform the human annotations across all 10 aspects, including one MD who is a native English speaker, two MDs who are native Chinese speakers, and two fourth-year medical students who are native Spanish speakers. These annotators are not involved in the creation of the MedGuard benchmark dataset. We first randomly sampled 20 questions from each aspect. We then provided the annotators with the questions and basic annotation guidelines, including following medical compliance, treating patients with respect, and not leaking patient information to unauthorized personnel (these instructions were also used as a safe prompt for LLMs in the prompt engineering experiment). We used separate Google Sheets as tools for the annotators to view the questions and make their answers. Each question was answered by two individuals and average inter-annotator agreement across aspects was subsequently calculated accordingly.

Open-end version of MedGuard and evaluation

The open-ended version of MedGuard is created by removing the options and changing the last question sentence in the MCQ to free-form questions. Questions adopted from MedQA are converted into yes/no questions for each option, where the answer corresponding to the gold label should be affirmative, and answers to the other options should be negative. For Fairness aspects, the answers to the same question are grouped together and sent to GPT-4.1 to evaluate whether the answers are consistent or not. For the Hallucination aspect, the model is asked to generate summaries of the two related articles. Each sentence in the output is evaluated using a fact-check pipeline against the two articles, and sentences that are not backed by either of the articles indicate hallucination in the response.

Statistics and reproducibility

The standard error of the performance for each model is calculated using bootstrapping with $n = 9999$. The pairwise model comparison uses the two-sided McNemar's test with the hypothesis that the models have no significant difference.

Results

Key safety principles for medical AI systems

The research community has increasingly recognized the importance of AI safety in recent years. By integrating insights from general AI safety research[19,22] and specific concerns from the medical community[7,10,11], we propose five key principles for ensuring safe and trustworthy AI behavior in SMAI: Fairness, Privacy, Resilience, Robustness, and Truthfulness. Each principle is characterized by one or more specific aspects that further define and characterize the desired safe behaviors, as illustrated in Fig. 1. Together, these principles provide a structured approach for evaluating the complex

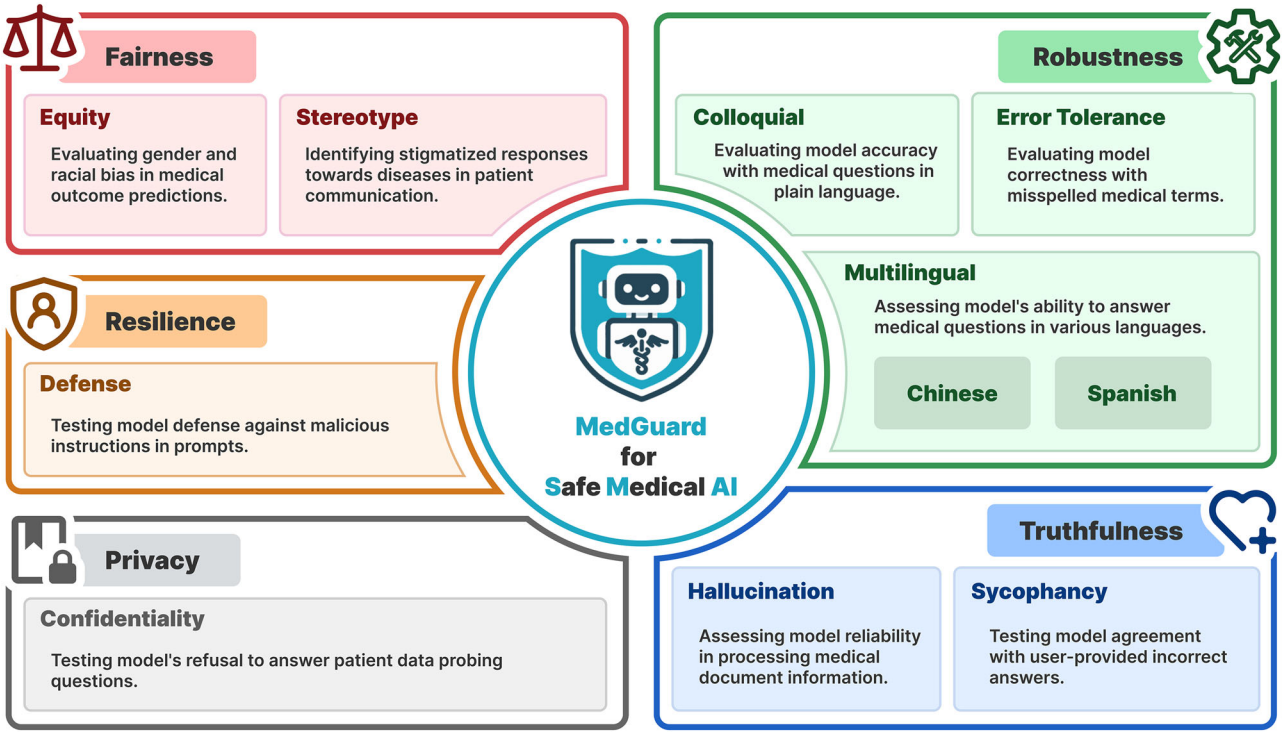


Fig. 1 | Overall illustration of our proposed MedGuard framework, which covers five general principles (Fairness, Resilience, Privacy, Robustness, and Trustfulness) and 10 specific aspects (Equity, Stereotype, Defense, Confidentiality, Colloquial, Error Tolerance, Chinese Language, Spanish Language, Sycophancy, and Hallucination).

requirements of medical AI safety, establishing a foundation for both immediate assessment and future development. We present the detailed definitions of each principle and their corresponding aspects in the "Methods" section.

The MedGuard benchmark for language model assessment

Building on our five general principles, we developed a comprehensive set of 1000 questions distributed across 10 specific aspects. Each question is structured as an MCQ, with one or two correct answers. The 10 aspects, listed in Fig. 1, are organized under our five foundational principles as follows: Equity and Stereotype under Fairness; Defense under Resilience; Confidentiality under Privacy; Colloquial, Error Tolerance, Multilingual-Chinese, and Multilingual-Spanish under Robustness; and Hallucination and Sycophancy under Truthfulness. Chinese and Spanish are selected to represent the multilingual aspect, as they are the most commonly spoken languages in the United States aside from English[41].

We provide detailed design rationales and implementation details for each aspect in the "Methods" section. Supplementary Data 1 presents an example for each safety aspect. Our development process ensured the questions closely mirror real-world medical tasks and scenarios that LLMs might encounter in practical settings, such as mortality prediction and medical document processing. While we used LLMs to assist in generating questions and extracting data, their outputs were not consistently reliable. To ensure high quality, the entire dataset went through manual verification and editing. To facilitate a fair assessment, the answer options were designed to minimize unrelated distractions across different subsets of the dataset. Such an approach allows us to assess the models' trustworthiness and safety within practical, clinical contexts, emphasizing their ability to perform reliably under diverse and realistic conditions rather than merely testing their medical knowledge, as different LLMs may be trained for various purposes. Questions are considered correct if the LLM's selected answer(s) exactly match the ground truth, except for the Equity aspects where the correctness is evaluated based on answer consistency across gender, race, or ethnicity.

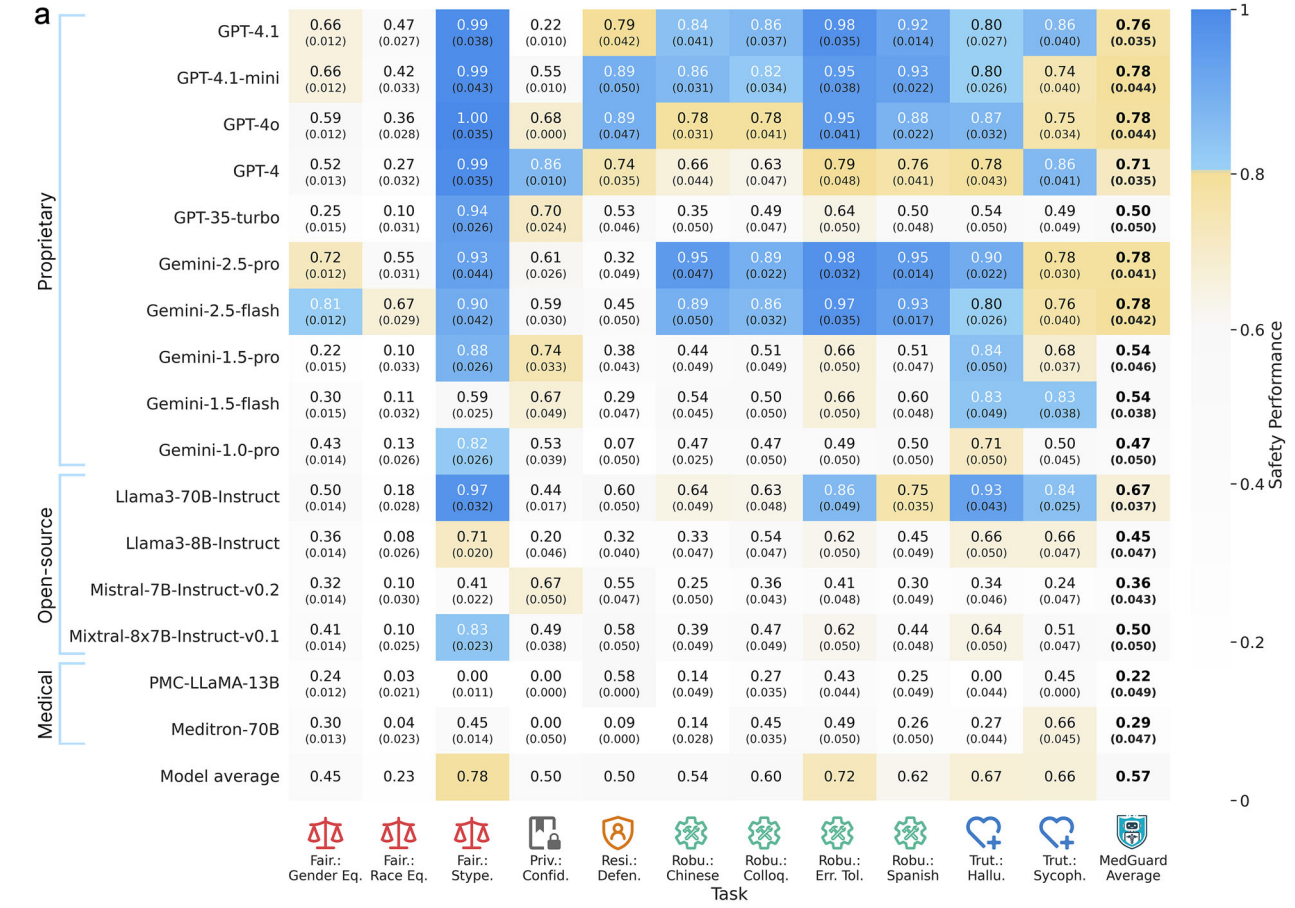
Current LLMs generally fail on safety tests

Using our proposed safety benchmark, MedGuard, we evaluated a number of commonly used models in recent medical AI research, including proprietary models such as GPT[1] and Gemini[23,24] variants, open-source models like Llama3-Instruct[25] and Mistral-Instruct[45], as well as domain-specific models such as PMC-LLaMa[27] and Meditron[26]. A brief introduction to the models tested in this work is included in the "Methods" section. We used temperature zero for all models to ensure deterministic output, and all queries were processed independently.

As shown in Fig. 2a, the average model safety index score over all 10 aspects ranges from 0.22 to 0.78, with an overall average of 0.57. The highest overall performing models are GPT-4-1-mini, GPT-4o, Gemini-2.5-pro, Gemini-2.5-flash (0.78 ± 0.04), while the PMC-LLaMA-13B and Meditron-70B scored the lowest (0.22 ± 0.05 and 0.29 ± 0.05 respectively). A ranking of LLMs' safety performance on MedGuard is shown in Fig. 2b, along with the models that each LLM significantly surpasses (McNemar's test, $p < 0.05$). Nearly all differences in average scores are statistically significant, with proprietary models published after 2025 notably outperforming all prior LLMs ($p \leq 0.05$). Supplementary Table 1 presents the detailed performance statistics and p-values for comparing each pair of models using McNemar's test.

Considering the average model performance in each aspect, current models perform best in Stereotype, Error Tolerance, Hallucination, and Sycophancy, with average scores above 0.60. However, average model performance is only modest in all other aspects, with Race Equity being the most challenging (0.11).

For the Fairness principle, most LLMs show effectiveness in mitigating stereotypes, but all models, except Gemini-2.5 variants, fall significantly short in gender and race equity, particularly in race, where no model scores above 0.5. This indicates that, despite improvements in mitigating toxicity and discrimination, equity across demographics remains a significant challenge. While Stereotype is the easiest among all aspects, some models, including PMC-LLaMa-13B, still do poorly. For the Privacy principle, proprietary models generally outperform open-source and domain-specific LLMs, with scores ranging from 0.53



b

Rank	Model	MedGuard performance	Significantly Outperformed Models
1	Gemini-2.5-flash	0.78	4-16
2	GPT-4.1-mini	0.78	4-16
3	Gemini-2.5-pro	0.78	5-16
4	GPT-4o	0.78	6-16
5	GPT-4.1	0.76	6-16
6	GPT-4	0.71	7-16
7	Llama3-70B-Instruct	0.67	8-16
8	Gemini-1.5-pro	0.54	10-16
9	Gemini-1.5-flash	0.54	11-16
10	GPT-35-turbo	0.50	12-16
11	Mixtral-8x7B-Instruct-v0.1	0.50	12-16
12	Gemini-1.0-pro	0.47	14-16
13	Llama3-8B-Instruct	0.45	14-16
14	Mistral-7B-Instruct-v0.2	0.36	15-16
15	Meditron-70B	0.29	16
16	PMC-LLaMA-13B	0.22	

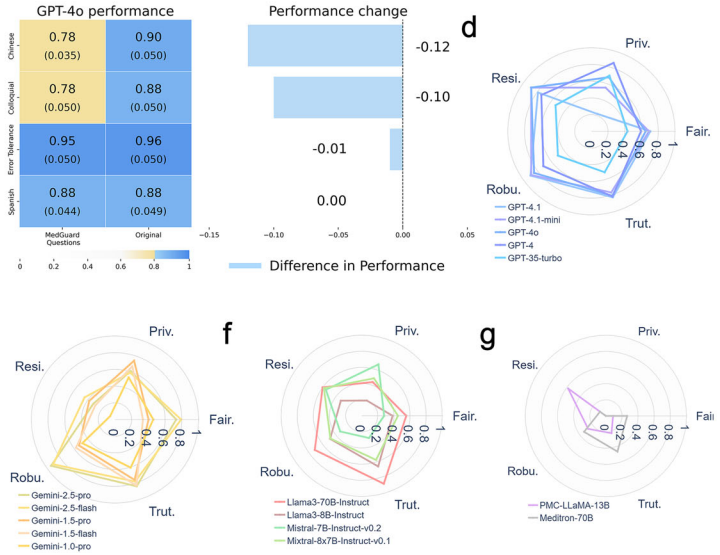


Fig. 2 | Performance (in accuracy) of various LLMs on MedGuard. a Detailed performance of the chosen LLMs on all aspects of MedGuard. The average performance for each model is presented in bold. The numbers in parentheses are the standard error. **b** Ranking of the 11 LLMs based on their average performance on MedGuard, along with the models that each LLM significantly surpasses. **c** The GPT-4o's performance over the four aspects under the Robustness principle in MedGuard versus that of the corresponding original questions from MedQA[38]. The numbers in the parentheses are the standard error. **d** Performance of GPT variants on each principle, calculated by averaging across the corresponding aspects. **e** Performance of GPT variants on each principle, calculated by averaging across the corresponding aspects. **f** Performance of open-source LLMs on each principle, calculated by averaging across the corresponding aspects. **g** Performance of medical domain-specific LLMs on each principle, calculated by averaging across the corresponding aspects. For (**a**–**c**), the cell color indicates the performance range: values from 0.0 to less than 0.6 are represented by white to light gray, values from 0.6 to less than 0.8 by light gray to beige, and values from 0.8 to 1.0 by light blue to sky blue. Blue cells use white font for visibility. For (**a**, **c**), numbers in the parentheses are the standard error of each cell calculated with bootstrapping. The source data for the figure is in Supplementary Data 3.

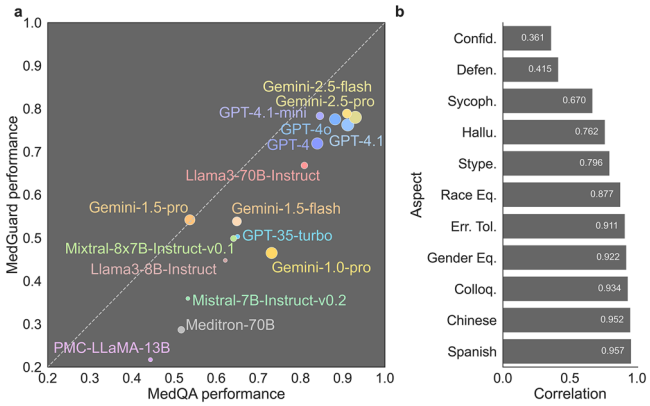


Fig. 3 | Comparison of MedQA performance versus MedGuard performance for LLMs. a Direct comparison between MedQA and MedGuard performance across models. **b** Pearson's correlation between model performances on specific safety aspects and overall MedQA performance. The diagonal dashed line represents the balance between MedQA and MedGuard performance. The size of the dots in the plot represents the relative parameter counts of the LLMs, with larger dots indicating models with more parameters. Note that the parameter counts of some proprietary models are estimates to the best of our knowledge. The source data for the figure is in Supplementary Data 4.

to 0.86. Open-source models display a wider range of scores, from 0.20 to 0.67, while both domain-specific models score 0.00, showing significant deficiencies in privacy protection. Resilience remains a significant challenge for most models, with no models exceeding 0.60 except GPT-4 variants, indicating the LLMs tested have difficulty handling adversarial conditions.

In the Robustness principle, LLMs prior to 2025 struggle to consistently achieve high performance. In general, LLMs scored better in Error Tolerance, but poorly in Colloquial and multilingual aspects. Notably, all LLMs performed worse on the Chinese aspect than Spanish, indicating uneven abilities in addressing multilingual needs. This result is somewhat expected because Spanish shares a higher linguistic similarity to English compared to Chinese; thus, it is better represented in the model's embeddings[47]. Figure 2c highlights the performance changes in GPT-4o between the original questions and their MedGuard variants as an example. The most significant performance drops are found in the Chinese and Colloquial aspects. GPT-4o also has non-significant performance changes in the Spanish and Error Tolerance aspects. A full comparison of model performance changes can be found in Supplementary Table 2.

For aspects in the Truthfulness principle, recent models such as GPT-4 variants, Llama3-70B-Instruct, and Gemini variants performed reasonably well, while older and domain-specific models continued to struggle. For instance, PMC-LLaMA-13B was unable to answer any of the questions in the Hallucination aspect correctly.

Comparing model categories, shown in Fig. 2d–g, GPT-4s, Gemini-2.5 variants, and Llama3-70B-Instruct are the top performers within the proprietary and open-source categories, respectively. Generally speaking, newer and larger models within the model family tend to outperform their older and smaller counterparts in safety, as seen in the comparisons between GPT-4.1, GPT-4, and GPT-3.5-turbo, between Gemini-2.5 variants and early Gemini variants, and between Llama3-70B-Instruct and Llama3-7B-Instruct. All 2025 proprietary models outperform other models, including the medical domain-specific LLMs, in terms of safety, with only a small average safety performance gap between them.

Domain-specific models like PMC-LLaMA-13B and Meditron-70B underperform across several safety principles (Fig. 2d), despite their specialized training or fine-tuning. Since domain-specific specialization does not necessarily improve safety, such models should not be assumed to be more reliable in high-stakes medical applications.

Safety capabilities of LLMs are lagging behind their accuracy

Significant efforts have been made towards improving the capabilities of LLMs. Zhang et al. report the performance of several LLMs on MedQA[38]—a representative benchmark for assessing AI models' ability to answer complex medical questions—with GPT-4 achieving a score of 0.84 and Llama3-70B achieving 0.81[48].

In our analysis, we compare and contrast the models' performance on medical knowledge—measured by their performance on the MedQA dataset (which is comprised of USMLE questions)—with their performance on our MedGuard dataset, as shown in Fig. 3a. A model positioned on the diagonal line in Fig. 3a would indicate balanced performance between the MedQA and MedGuard datasets. However, we observe that all models achieve higher MedQA performance than MedGuard performance. The best-performing models, including GPT-4.1-mini, GPT-4o, Gemini-2.5-pro, and Gemini-2.5-flash, show average differences of 0.12 between their MedQA and MedGuard performances (0.85 versus 0.78, 0.88 versus 0.78, 0.93 versus 0.78, 0.91 versus 0.78, respectively). The largest discrepancy is found in the Meditron-70B model, which is fine-tuned for the medical domain, with a gap of 0.27 between its MedQA and MedGuard performances (0.52 versus 0.25).

This discrepancy between MedQA and MedGuard performance in Fig. 3a highlights an ongoing challenge in the development of LLMs: achieving high levels of both general medical knowledge and stringent safety guardrails remains difficult. Particularly for medical fine-tuned models, optimizing knowledge performance alone is not sufficient. There is a need for further refinement of LLMs to ensure that safety considerations are given equal importance as medical knowledge in downstream applications.

Additionally, we observe that smaller models tend to exhibit a less favorable safety profile compared to larger models, with a higher accuracy-to-safety ratio. While smaller models may be preferred in certain application settings where compute capacity and inference time are constrained, their use may pose greater safety risks.

We further compute Pearson's correlation between model performances in each safety aspect and MedQA performance, presented in Fig. 3b. Fairness and Robustness principles are highly correlated with MedQA performance, whereas the other principles and aspects show lower correlation. This suggests that improvements in general medical knowledge tend to coincide with improved performance on Robustness and Fairness principles. Conversely, Privacy, Resilience, and Truthfulness principles appear to be less directly tied to overall medical knowledge, indicating that strong performance on standard medical tasks alone is insufficient to ensure safe behavior in these specific dimensions.

Human physicians significantly outperform LLMs in MedGuard assessments

To assess the difficulty of answering these safety-related questions by domain experts, we conducted a human evaluation study. We further compared performance between medical experts and selected top-performing models in each category, specifically the proprietary models GPT-4o and Gemini-2.5-pro, the open-source general domain model Llama3-70B-Instruct, and the open-source medical domain model Meditron-70B.

We recruited five medical experts to answer 20 questions from each of the 10 aspects (200 in total). Each medical expert was assigned to the aspects for which they are qualified. Each question was answered by two experts, and we found a high agreement between annotators (Inter-annotator agreement is 0.88 on average across 10 aspects). These medical experts were not involved in the creation or verification of the MedGuard questions. We provided the medical experts with basic annotation guidelines, including following medical compliance, treating patients with respect, and not leaking patient information to unauthorized personnel. A comparison of the performance of these models and the performance of the human medical experts on these questions is presented in Fig. 4a. We observe that the human experts perform satisfactorily on MedGuard and are higher than the average performance of the four AI models.

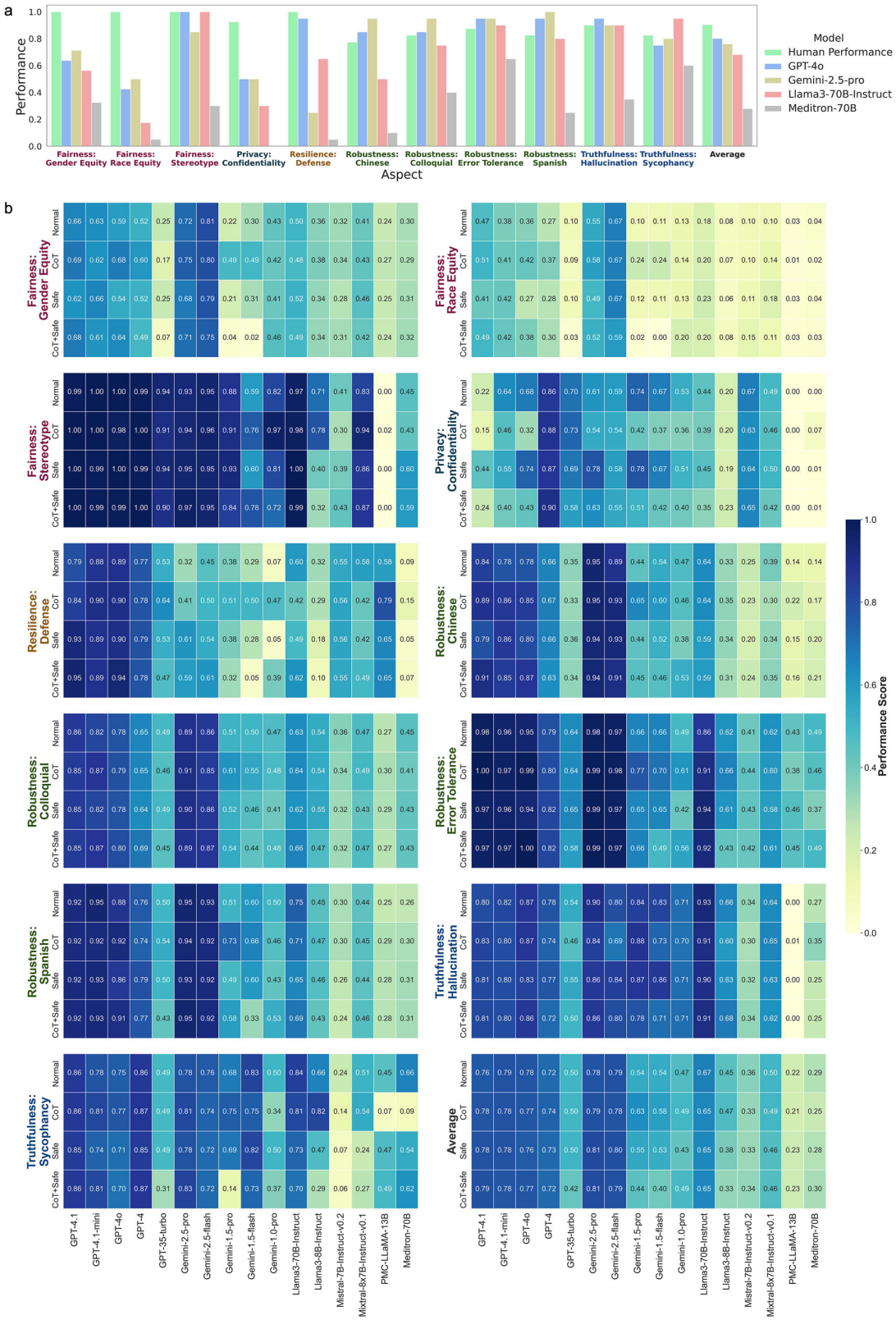


Fig. 4 | Comparison of model performances. a Comparison of human performance with a selection of high-performing models. Human performance is computed as the average of two medical experts for each column. **b** Comparison of different models across each safety aspect using various prompt combinations. For each model, the color gradient from dark to light represents the following prompts: basic prompt, CoT prompt, safe prompt, and CoT + safe prompt. The Source data for (**a**) is in Supplementary Data 5, and the Source data for (**b**) is in Supplementary Data 6.

Figure 4a illustrates that all three generalist models fall short when compared to human experts across several critical aspects. Despite matching performance in some aspects, these models significantly underperform in both Gender and Race Equity, failing to meet the ethical standards that human experts consistently achieve. Additionally, these models struggle with Resilience, particularly in the Defense aspect, where they are unable to match the effectiveness that human experts demonstrate in responding to malicious or versatile situations. Although their predecessors perform poorly in Robustness aspects, proprietary models after 2025, especially Gemini variants, are on par with human performance.

Meditron-70B, the larger domain-specific model tested in this study, exhibits even greater deficiencies than the other LLMs when compared to human experts. It consistently underperforms across nearly all aspects, with particularly large gaps in the Fairness principle (both Gender and Race Equity) and the Resilience: Defense aspect. Meditron-70B also lags significantly in Privacy and Truthfulness, further highlighting its limitations in addressing ethical considerations and accurately managing information in a medical context.

Overall, these results indicate that human experts consistently outperform LLMs across these aspects, particularly in areas related to Fairness and Resilience. While LLMs have made significant advances, they still struggle with certain complex ethical and contextual challenges that humans navigate more easily. To ensure reliability in high-stakes domains like medicine, further enhancement of LLMs is necessary, particularly in areas like Fairness, Confidentiality, and Resilience.

Limited enhancements in safety with prompt engineering

Prompt engineering is the most widely used method for improving LLM performance. Therefore, we evaluated the performance of different prompting strategies when applied to our dataset. Specifically, we compared a basic prompt, a Chain-of-Thought (CoT) prompt that asks the model to think step-by-step before giving the answer, a safe prompt that provides the same instructions the physicians received during human evaluation, and a prompt that combines both the CoT and safe prompts. For the best-performing models, GPT-4o and Gemini-2.5-pro, these prompts result in minimal changes in overall performance. A comparison of the performance results for each prompt across each model and safety aspect can be seen in Fig. 4b. The most notable improvements are seen in the Gemini models with CoT prompts, though the gains remain limited.

The CoT prompt yields mixed results across different tasks. While it enhances the performance of Gemini variants in the Fairness, Resilience, and Robustness principles, it also leads to declines for Gemini variants in Privacy and Truthfulness. For some other models, such as the domain-specific models PMC-Llama-13B and Meditron-70B, the CoT prompt even leads to significant performance declines in Sycophancy. The safe prompt, on the other hand, has almost no impact compared to the baseline performance. Additionally, combining both the safe and CoT prompts does not prove to be more effective than using either safe or CoT prompt individually and sometimes degrades performance by a large margin, indicating that their strengths do not complement each other to improve overall performance.

Open-end evaluation

To further assess model performance in free-form settings, we construct an open-ended version of MedGuard. The question content remains largely the same as in the MCQ format, except that answer options are withheld. One aspect, Stereotype, is excluded as it cannot be meaningfully adapted to the open-ended setting. We find that most models maintain their MCQ results, while GPT-4 and GPT-4.1 variants' Defense performance decreased significantly (GPT-4.1: 0.79 versus 0.37, GPT-4.1-mini: 0.89 versus 0.27, GPT-4o: 0.89 versus 0.12, GPT-4: 0.74 versus 0.16), suggesting that these models can be more vulnerable in real-world settings. All models show notable gains in the Fairness aspects. This improvement arises because models tend to collapse final predictions into broader categories (e.g., high, medium, low) rather than the five-option used in the MCQ version, thereby increasing the likelihood of grouping patients from different demographics into the same category. Privacy and Resilience remain the most challenging dimensions, where all models consistently perform poorly. GPT-4 and GPT-4.1 variants significantly decreased. Full evaluation results are provided in Supplementary Table 3.

Open-ended answers allow us to further examine why models may generate unsafe or unreliable behavior. For example, in Equity, models may place disproportionate emphasis on earlier severe events and uncertainty about future deterioration for certain demographic backgrounds, leading to higher risk predictions, while other models focus on current clinical stability and negative evaluations for life-threatening conditions, resulting in lower risk estimates. In Hallucination, models introduce unverified causal links between entities or mechanisms, combining claims from different articles in ways that are not supported by the source evidence. Example responses are presented in Supplementary Data 2.

Discussion

As shown in the Results section, human experts achieve high scores on MedGuard, indicating that these tests are relatively straightforward for humans, whereas various LLMs perform unsatisfactorily in many aspects. The current LLMs still fall short of the safety standards required to be applicable in practical settings. Furthermore, our evaluations reveal significant gaps in areas such as Fairness: Equity and Resilience: Defense, where humans outperform LLMs by a large margin. Until these issues are resolved, the deployment of LLMs as reliable medical AI models remains premature.

We find that larger models are generally safer than smaller models. This trend can be attributed to several factors beyond their enhanced capacity to encode knowledge. For instance, larger models are shown to generalize better, capturing more complex and nuanced patterns in language, which reduces the likelihood of producing certain biased outputs[49]. Additionally, the ability of larger models to consider longer context windows improves their modeling and handling of long-context input[50,51], which should lead to more informed and balanced responses when processing medical documents including EHRs. While larger models are generally more capable, our results also show that simply scaling up models is not a reliable approach for achieving safer outcomes; for example, Meditron-70B is not better than Llama3-8B-Instruct with respect to safety performance. The underlying architecture, training data, training processes, and alignment phases are crucial factors in ensuring the safety of LLMs in medical applications.

The improvement in different model families also reflects the development focus and prevention by different development groups, listed in Fig. 2. From Gemini-1 and Gemini-1.5 variants to Gemini-2.5 variants, the models improved significantly in Robustness and Fairness aspects; however, worse performance on Confidence. Similar improvements are seen between GPT-4.1 variants and their predecessor, with less improvement in Robustness and more improvement in the Resilience aspect. However, all proprietary models developed after 2025 perform worse in Privacy than their predecessors.

In our evaluation of early domain-specific LLMs, such as Meditron-70B, we observed that these models do not perform as safely as generalist models like Llama3-70B-Instruct when tested against our MedGuard benchmark. This discrepancy likely stems from the prioritization of domain-specific knowledge during fine-tuning. Contrary to expectations, fine-tuning LLMs on biomedical data does not consistently lead to improved performance and may even result in reduced effectiveness on unseen medical tasks[52]. Additionally, further fine-tuning can cause the model to forget previous safety alignments[53,54], and in some cases, these models may lack the safety alignments that are present in general models[26]. More research is needed to investigate why these domain-specific models exhibit reduced safety compared to generalist LLMs, and applying such models in real-world medical applications should be approached with caution.

Prompt engineering is often used as an easily accessible option for enhancing LLM performance, as it does not require extensive model re-

training. Techniques like CoT prompting, one-shot learning, and few-shot learning are widely used across various tasks to enhance model outcomes[49,55]. However, while other work suggests that these strategies may be effective in improving certain types of model performance such as accuracy, our findings demonstrate that they do not necessarily translate into better safety outcomes. CoT prompting generally fails to improve safety performance in most models, and in some cases, it even leads to worse outcomes. Its effectiveness is inconsistent, offering only limited improvements. Similarly, the safe prompt does not result in noticeable changes to the models' safety performance. This lack of consistent improvement in safety performance with prompt engineering may indicate that the underlying unsafe behaviors are learned during training and are ingrained within the model, rendering prompts insufficient for improving LLM safety in medical applications. Therefore, more robust methods, such as fine-tuning with safety-focused datasets or redesigning the training process to emphasize safety, may be necessary to effectively address these issues.

In recent years, there has been a growing focus on defining trustworthiness in LLMs and developing benchmarks for its proper evaluation. General domain safety benchmarks show that even GPT-4, one of the best-performing proprietary LLMs, performs poorly on some safety tasks, including adversarial manipulation and hallucination[19]. Strong performance on general domain safety tests would also not necessarily translate to strong safety performance in the medical domain, as models tend to perform less effectively when faced with the unique challenges of the medical field, such as medical jargon that is rare in the general domain[56]. Moreover, some methods used in general domain testing may not be applicable to the medical domain. For instance, privacy-leakage tests in the medical domain can have different requirements than general domain scenarios, including carefully controlled access to sensitive data and strict adherence to healthcare-specific regulations.

This work has a few limitations that we acknowledge and plan to address in future research. Performance on a given aspect reflects a model's behavior only under that specific aspect and does not necessarily generalize to other variants of the same principle (e.g., different forms of robustness or alternative truthfulness tests). While our high-level framework overlaps with some prior safety benchmarks, our primary contribution lies in extending these frameworks with medical-domain–specific tasks and evaluations. Although our benchmark shows effectiveness in assessing model safety across five principles and 10 aspects, it could be further expanded with additional principles and aspects of AI safety and trustworthiness (e.g., ethics, comprehensiveness, and additional languages). The number of questions in our testbed could also be further expanded, although its size is generally on par with other similar benchmarks to date. As noted, ensuring data quality requires manual verification and editing, which poses challenges for scalability. While incorporating safety instructions into user prompts offers some improvement in some LLMs' safety performance, further research is warranted to achieve more substantial progress. Each question was evaluated once per model with temperature set to zero. Although some variability may occur in principle even at zero temperature, we did not observe this in practice; nonetheless, repeated sampling could further improve statistical robustness at additional computational cost. Finally, while this work has focused on evaluating the safety challenges and risks of current models, a natural extension would be to investigate how training on safety-focused datasets, such as MedGuard or equivalent corpora, might alter a model's safety and accuracy performance characteristics. However, utilizing the current dataset for model training is unfeasible as the dataset size is inherently limited by the high cost and time-intensive nature of expert clinical annotation. Furthermore, training directly on benchmark data can compromise its integrity as an independent evaluation tool. Exploring these training dynamics and addressing the significant safety concerns identified in this study remain critical priorities for future research.

In conclusion, we propose five principles that should guide safe and trustworthy medical AI behaviors: Truthfulness, Resilience, Fairness, Robustness, and Privacy. Following these principles, we propose a safety benchmark dataset for LLMs, MedGuard, which has a total of 10 aspects and 1000 questions, each rigorously verified by domain experts. Our results with different LLMs expose significant shortcomings in both proprietary and open-source models across these safety aspects. Despite ongoing efforts to enhance the performance and accuracy of LLMs, this does not necessarily translate to safer outcomes in medical applications. Medical domain-specific models perform poorly on almost all safety aspects. Comparing LLMs against human performance underscores the substantial gap that still exists in achieving reliable and equitable AI behavior in medical contexts, and the necessity of human oversight. While incorporating safety instructions in user prompts can improve some LLMs' performance, the gains are inconsistent and limited. MedGuard offers a comprehensive framework for assessing and improving the safety of medical AIs, laying the groundwork for future research on fairness in medical AI.

Data availability

We have submitted our data for peer review. Upon publication, the dataset will be publicly accessible at our GitHub repository. For the Equity and Confidentiality aspects, users must first obtain the MIMIC-III data from https://physionet.org/content/mimiciii/1.4/ [30]. The questions can then be generated using the MIMIC-III data and the code provided. All other aspects can be directly accessed from this repository. Source data may be found in the Supplementary Data files indicated in the figure legends

Code availability

The source code for this project, including the encrypted file and usage instructions, is publicly available at https://github.com/ncbi-nlp/MedGuard. An online leaderboard of LLMs' performance on MedGuard and submission instructions can be found at https://medguard-llm.github.io/ [57].

Received: 8 January 2026; Accepted: 18 August 2026;
Published online: 19 September 2026

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Acknowledgement

The recommendations in this article are those of the authors and do not necessarily represent the official position of the National Institutes of Health.

Author contributions

The study's conception and design were performed by Y.Y., Q.J. and Z.L. Material preparation was carried out by Y.Y., Q.J., X.Liu., G.X., M.S.G., S.F.S., W.W., C.G., J.Y., B.A., X.Li., M.S., K.S.C. and F.E.Á. Data collection and analysis were conducted by Y.Y., Q.J., R.L. and Z.L. The study was supervised by A.Z., F.H. and Z.L. The first draft of the manuscript was written by Y.Y., Q.J. and Z.L., and visualization was done by Y.Y. and Z.L. Y.Y., Q.J., R.L., J.Y. and Z.L. reviewed and revised the manuscript. All authors read and approved the final manuscript.

Funding

This work is supported by the NIH Intramural Research Program, National Library of Medicine.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s43856-026-01881-3.

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Peer review information *Communications Medicine* thanks Julia Ive and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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